

Objectives

Paragraph

The objective of Day 18 is to understand and implement **exception handling in Python**. The project focuses on using try, except, else, and finally blocks to safely handle expected runtime problems such as invalid input, division by zero, invalid list indexes, and missing dictionary keys. The Streamlit interface makes it possible to test different error conditions interactively.

Bullet Points

- Understand Python exception handling.
- Practice try, except, else, and finally.
- Handle invalid numeric input.
- Handle division by zero.
- Handle invalid list indexes.
- Handle missing dictionary keys.
- Display meaningful error messages.
- Prevent the application from crashing.

Technologies Used

Tools

- Python
- Streamlit
- VS Code / Python IDE
- Git
- GitHub

Deliverables

- Exception-handling demonstration program
- At least five handled error cases
- Interactive Streamlit application

- Meaningful error messages
- Examples of successful and failed operations
- Explanation of common Python exceptions

C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY18>

```
import streamlit as st

st.set_page_config(
    page_title="Exception Handling Practice",
    page_icon="⚠️",
    layout="centered"
)

st.title("⚠️ Exception Handling Practice")
st.write("Practice handling common Python exceptions using Streamlit.")
st.divider()

st.subheader("1. Numeric Input Validation")

number_input = st.text_input(
    "Enter a number:",
    placeholder="Example: 25"
)

if st.button("Check Number"):
    try:
        number = int(number_input)
        if number < 0:
            raise ValueError("Negative numbers are not allowed.")
    except ValueError as e:
        st.error(f"❌ Value Error: {e}")
    else:
        st.success(f"✅ Valid number: {number}")
    finally:
        st.info("Operation completed.")

st.divider()

st.subheader("2. Division Calculator")

col1, col2 = st.columns(2)

with col1:
    first_number = st.number_input(
        "First Number",
        value=10.0
    )

with col2:
    second_number = st.number_input(
        "Second Number",
        value=2.0
    )

if st.button("Divide"):
    try:
        result = first_number / second_number
    except ZeroDivisionError:
        st.error("❌ Cannot divide by zero.")
    except TypeError:
        st.error("❌ Invalid data type.")
    else:
        st.success(f"✅ Result: {result:.2f}")
    finally:
        st.info("Division operation completed.")

st.divider()

st.subheader("3. List Access")

items = ["Python", "Streamlit", "Git", "GitHub"]

index = st.number_input(
    "Enter an index (0-3):",
    min_value=0,
    max_value=10,
    value=0,

```

Output:- <http://localhost:8501>

⚠️ Exception Handling Practice

Practice handling common Python exceptions using Streamlit.

1. Numeric Input Validation

Enter a number:

78

Check Number

2. Division Calculator

First Number

76.00

- +

Second Number

65.00

- +

Divide

3. List Access

Enter an index (0-3):

Access Item

4. Dictionary Key Handling

Enter a key:

Find Key

✖ Key Error: 'trishika' does not exist.

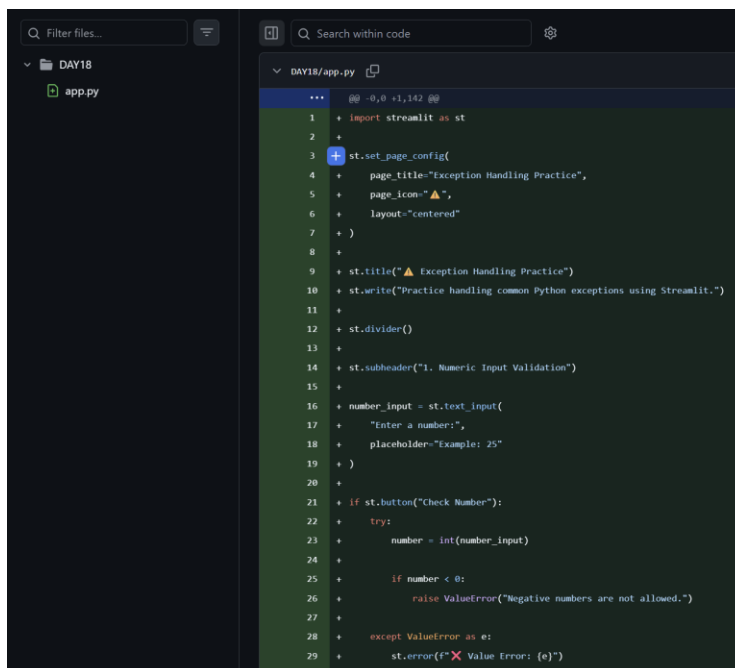
Dictionary operation completed.

Exceptions Covered ↔

- **ValueError** – Invalid numeric or value input
- **ZeroDivisionError** – Division by zero
- **IndexError** – Accessing an invalid list index
- **KeyError** – Accessing a missing dictionary key
- **TypeError** – Invalid data type operation
- **finally** – Executes whether an exception occurs or not
- **else** – Executes when no exception occurs

🎉 Exception handling practice completed!

Github: <https://github.com/shambhavi0804/VEDA-internship/commit/55829e6c920f225e0cda106483fae7cdbf1e2298>



```
... @ -0,0 +1,142 @@
1 + import streamlit as st
2 +
3 + st.set_page_config(
4 +     page_title="Exception Handling Practice",
5 +     page_icon="🚨",
6 +     layout="centered"
7 + )
8 +
9 + st.title("🚨 Exception Handling Practice")
10 + st.write("Practice handling common Python exceptions using Streamlit.")
11 +
12 + st.divider()
13 +
14 + st.subheader("1. Numeric Input Validation")
15 +
16 + number_input = st.text_input(
17 +     "Enter a number!",
18 +     placeholder="Example: 25"
19 + )
20 +
21 + if st.button("Check Number"):
22 +     try:
23 +         number = int(number_input)
24 +
25 +         if number < 0:
26 +             raise ValueError("Negative numbers are not allowed.")
27 +
28 +     except ValueError as e:
29 +         st.error(f"✖ Value Error: {e}")
30 +
```

Linkedin:-<https://lnkd.in/p/ddM8Nq52>



Exception Handling Practice

Practice handling common Python exceptions using Streamlit.

1. Numeric Input Validation

Enter a number:

78

Check Number

2. Division Calculator

First Number

76.00

-

+

Second Number

65.00

-

+

Divide



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 GIF



VEDA Internship – Day 18 Completed! Today, I worked on Task 18: Practice Exception Handling as part of my Python Programming Internship. I built an interactive Exception Handling application using Python and Streamlit to understand how applications can... more